Dear Editors and Reviewers,

We appreciate the thorough review of our manuscript and valuable suggestions. Below, we present our point-by-point responses and describe the corresponding changes incorporated into the revised version of the manuscript.

**1. Concern 1: Motivations of Strong Blocking Flows Condition.**

We agree that the previous draft did not sufficiently address the motivation of the strong blocking flows condition. We have clarified in Section 3.3 that this is an additional restriction on the presence or absence of hyperarcs, rather than their numerical values. It does not follow from stationarity or an aggregate balance condition.

We have added backward dynamical neighbors $Q_A$, defined as inclusion-minimal sets satisfying $H_A|_{A' \cup Q'_A} = 0$, alongside the original dynamical neighbors $K_A$. The reversed-transfer-entropy hypergraph is now defined using $Q_A$. Under deterministic reversible dynamics, both minimal sets exist. Their union $K_v^{\text{both}} = K_v \cup Q_v$ gives $\alpha_1^{K_v^{\text{both}} \to v} = 0$. Section 3.2 uses the separate expansions

$$\Delta H_v = I_{(K_v):v'}|_v - I_{(Q'_v):v}|_{v'},$$

so that all terms needed for the individual entropy dynamics are represented. No claim of completeness for all multiple-mutual-information dynamics is made.

The copying example in Section 3.3 has been completed as a closed three-component system:

$$\Gamma_v(t + \Delta t) = \Gamma_u(t), \quad \Gamma_u(t + \Delta t) = \Gamma_w(t), \quad \Gamma_w(t + \Delta t) = \Gamma_v(t).$$

With independent fair binary states, this dynamics is deterministic and reversible, and the uniform distribution is invariant. Its transfer entropies form a directed cycle, and its reversed transfer entropies have the opposite directions. All represented quantities equal $\ln 2$. Thus, the strong blocking flows condition and the combined-set $\alpha_1 = 0$ condition hold.

We acknowledge that the stronger condition restricts the applicable class of dynamical systems. Section 3.3 also gives a reversible controlled update with nonzero forward and reversed transfer entropy in the same direction, which violates blocking. The stochastic examples in Appendix C are retained as local illustrations of the focal updates, rather than realizations of the reversible assumptions for the whole system. The missing nonzero inverse-XOR channels and their reversed counterparts have been included.

**2. Concern 2: Unrealizable Cases in Lemma 5.1.**

We acknowledge the structural constraint pointed out by the reviewer. If a vertex has no B-inflow, Lemma 4.1 and the dynamical-neighbor definition give $H_{v'}|_v = 0$. Every multiple transfer entropy whose target contains $v$ then vanishes. Hence, an F-inflow without a B-inflow is impossible, and those cases have been removed from Lemma 5.1.

With separate forward and backward neighbors, Proposition 4.2 now states that a B-inflow is present exactly when $H_{v'}|_v > 0$, and an F-outflow is present exactly when $H_v|_{v'} > 0$. Every non-isolated vertex has at least one of these flow types. An F-inflow requires a B-inflow, and a B-outflow requires an F-outflow. Proposition 4.3 treats both BF-sinks and BF-sources: their entropy changes are strictly positive and strictly negative, respectively. Stationarity requires both a B-inflow and an F-outflow, together with numerical balance of the two aggregate contributions.

The cycle-necessity statement now specifies a fixed nonempty flow topology and assumptions holding at every step. It requires the existence of a weak BF-hypercycle, without claiming that every vertex belongs to a cycle. A reversible swap illustrates why the qualification concerning the temporal topology is necessary.

**3. Changes of Notations.**

We have modified notations about information-theoretic quantities by defining them with sets. This enables more compact descriptions and highlights the inclusion–exclusion principle of multiple mutual information (co-information). We have also distinguished backward neighbors $Q_A$ from the forward-neighbor layers in Appendix A. The counterexample in Appendix A and the signs in the conditioning expansions have been corrected. Unresolved figure references, the incomplete expression, the inverse-map notation mismatch, and appendix equation numbering have been corrected.

**4. Update of Reference.**

We have revised Ref. [25] to the latest preprint version.